

POLITECNICO
MILANO 1863

DEPARTMENT OF MECHANICAL
ENGINEERING

EXERCISE CLASS 4

Violation of Assumptions

Name Surname



DIPARTIMENTO DI ECCELLENZA
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Milano

Exercise 1

A chemical plant operates several reactors with low temperature solvents, where temperature fluctuations can affect reaction rates, product yield, and safety. To ensure process stability, the quality control team conducts regular temperature sampling throughout the facility. At fixed time intervals, temperature readings are taken from four randomly selected reactors or processing units. The collected data in °C is reported in *reactors.csv*.

- 1. Design a control chart for the mean.**
- 2. Consider the samples as individual measurements. Design a control chart for the mean.**

Exercise 1.1 Design a control chart for the mean.

1) Import the data in Minitab:

File → Open → Locate “reactors.csv” and Open

2) Inspect the dataset by plotting individual data points:

First, let’s stack the data into a single column for further analysis:

Data → Stack → Rows → Insert columns x1 x2 x3 x4 and store row subscripts

Let’s also create a column for column subscripts:

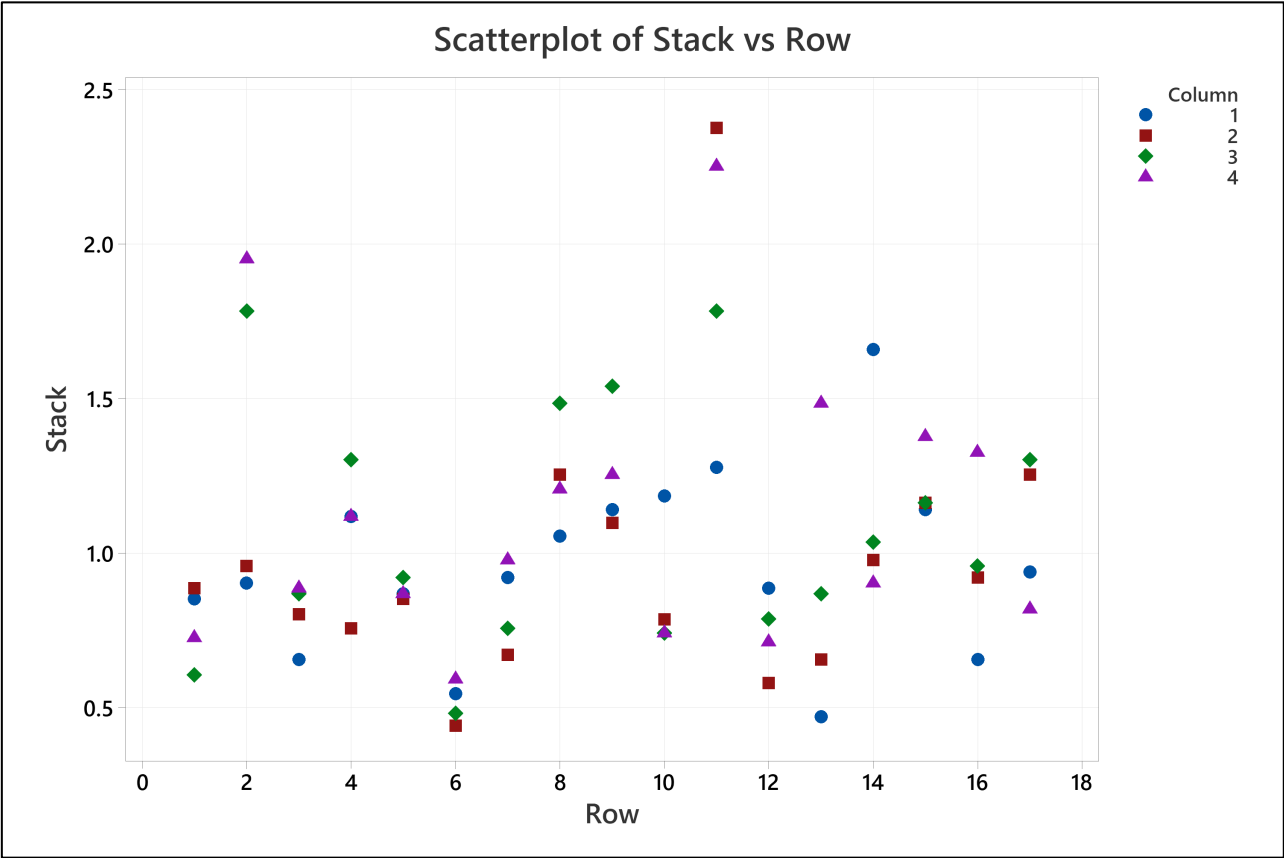
Calc → Make Patterned Data → Simple Set of Numbers → repeat 17 times the sequence 1,2,3,4

To create a scatterplot:

Graph → Scatterplot → Simple → set the stack as Y and the row subscripts as X

Let’s edit the scatterplot to add group attributes:

Double click on the graph → Click on a datapoint → In the Group tab add the patterned data as Categorical Variables for Grouping



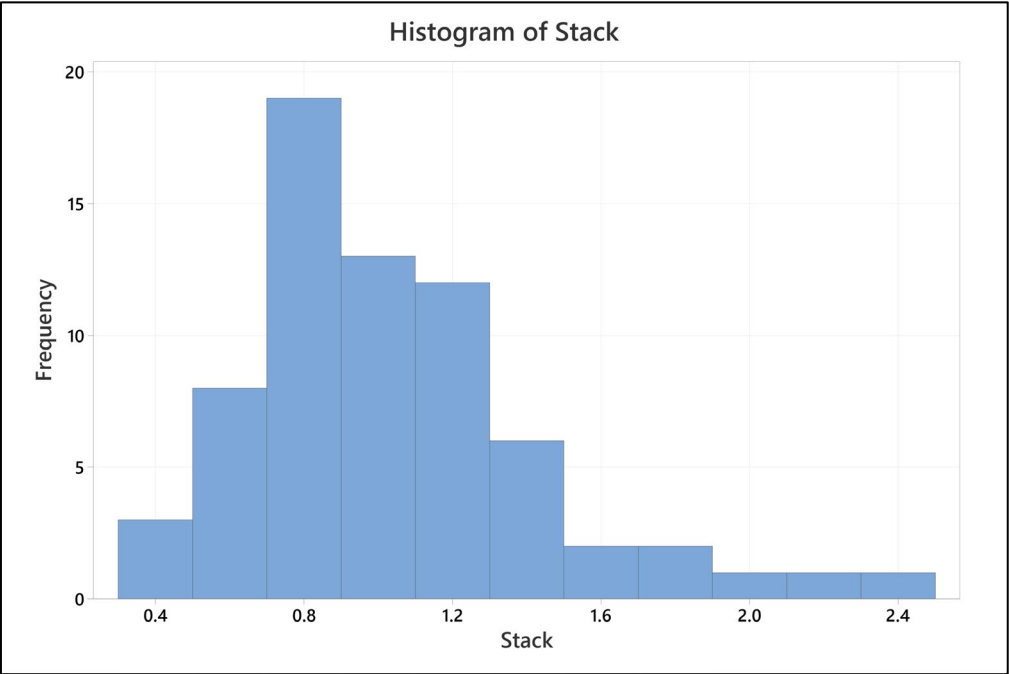
Looks like outliers are present, or - more likely - the distribution is skewed.

Exercise 1.1 Design a control chart for the mean.

3) Inspect the dataset by plotting the histogram of the data stack:

Graph → Histogram → Simple → Select the data stack

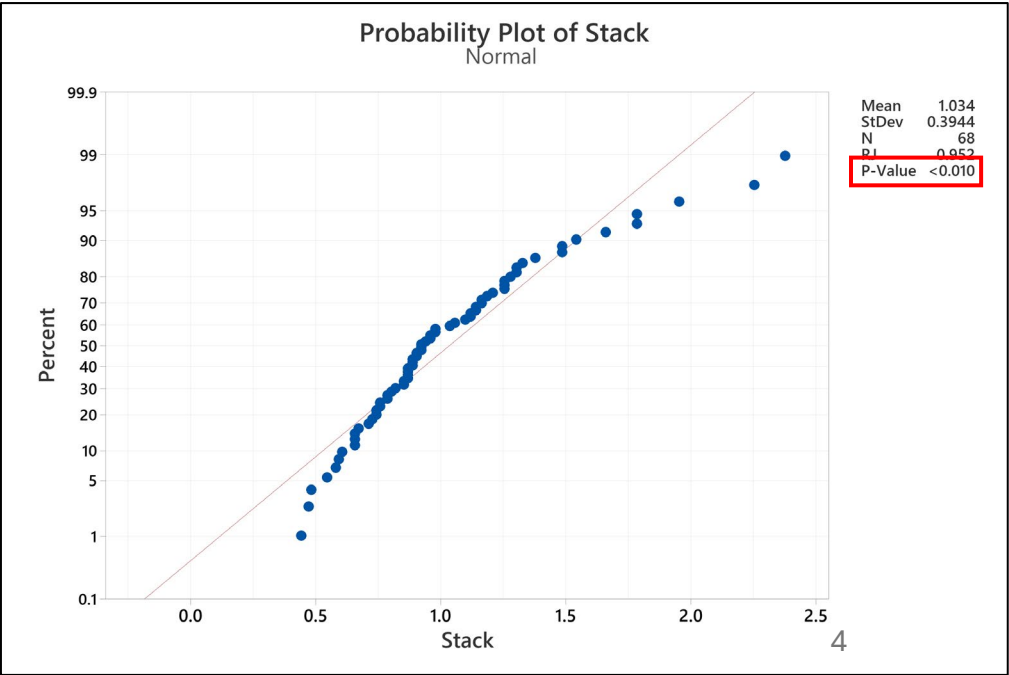
The distribution looks skewed. Let’s verify the normality assumption.



4) Normality assumption :

Stat → Basic Statistics → Normality Test → Select the data stack and Shapiro-Wilk test

The data are non-normal. Therefore, we cannot use the X-bar and R chart on the raw data. We need to transform the data. But what happens if we neglect the normality violation and use the X-bar and R chart on the raw data?

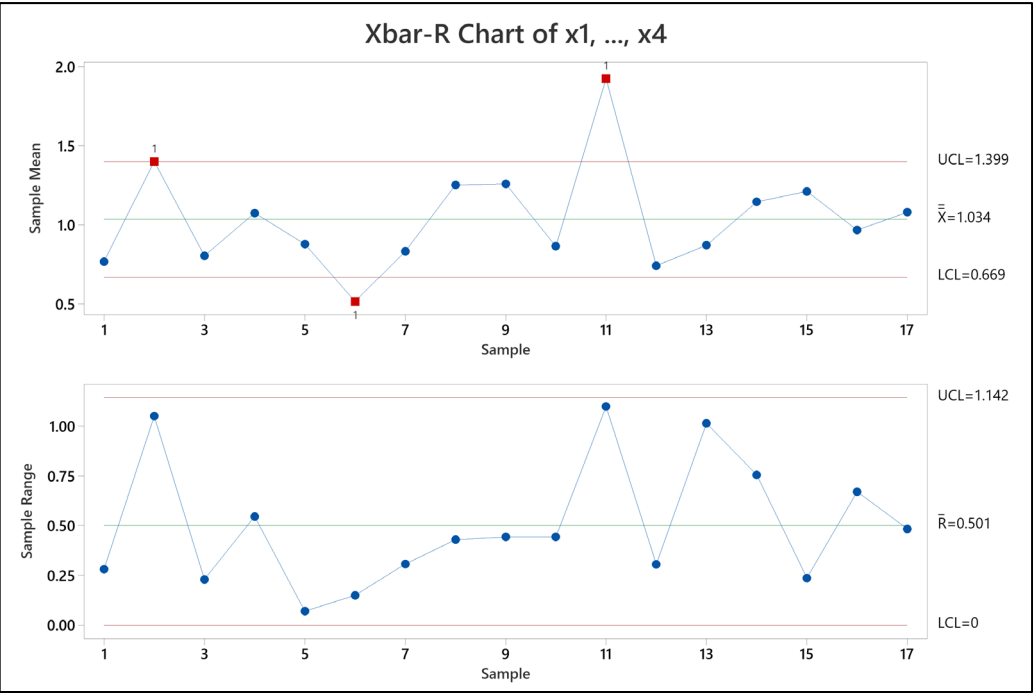


Exercise 1.1 Design a control chart for the mean.

5) Xbar-R chart on raw data (the normality assumption is violated) :

Stats → Control Charts → Variable Charts for Subgroups → Xbar-R

There are a few out-of-control observations, which may be due to a violation of the normality assumption. If the data is not normally distributed, the computation of control limits may not be valid.

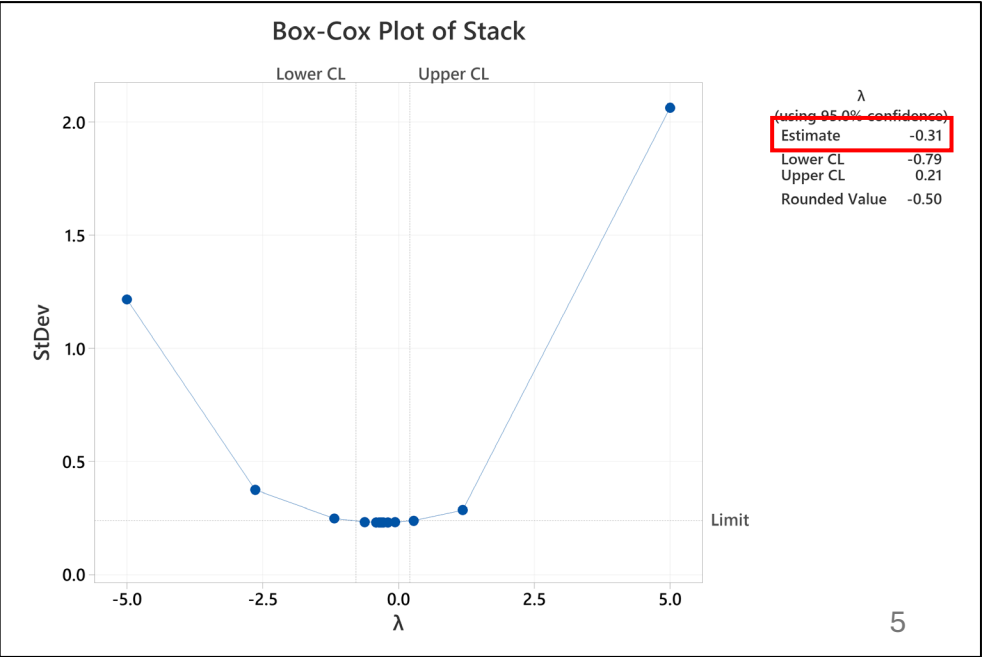


6) Let's transform the data using the Box-Cox transformation:

$$x_{BC,i} = \begin{cases} \frac{x_i^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln x_i & \text{if } \lambda = 0 \end{cases}$$

Stats → Control Charts → Box-Cox Transformation → All observations for a chart are in one column → Set the subgroup sizes at 4

By default, the Box-Cox function used Lambda = -0.31. A more interpretable (and close to optimum) value is Lambda = 0.

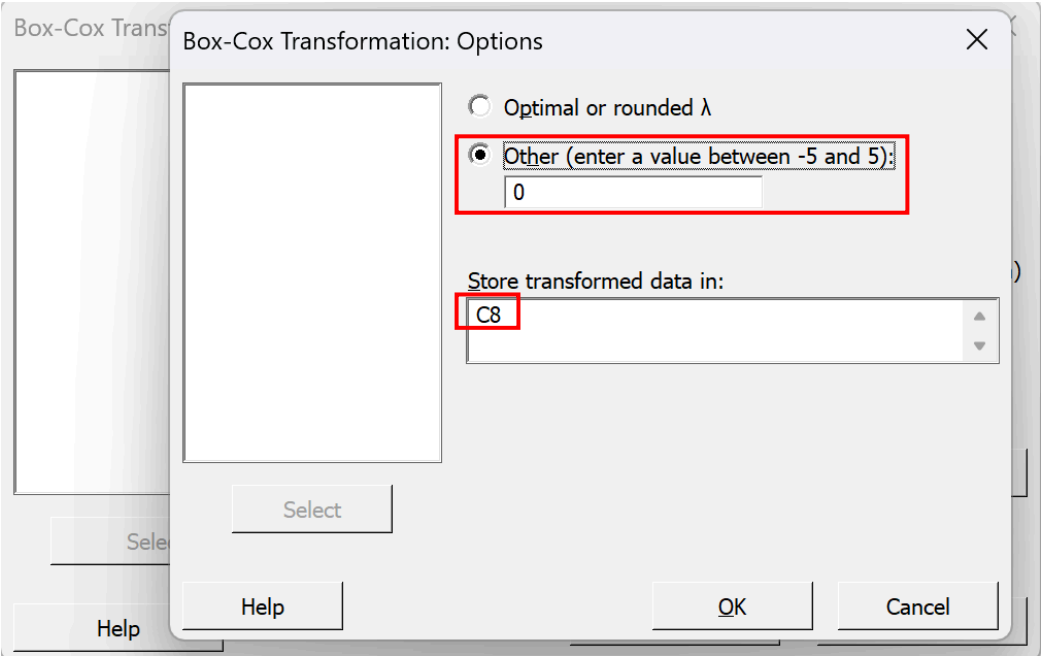


Exercise 1.1 Design a control chart for the mean.

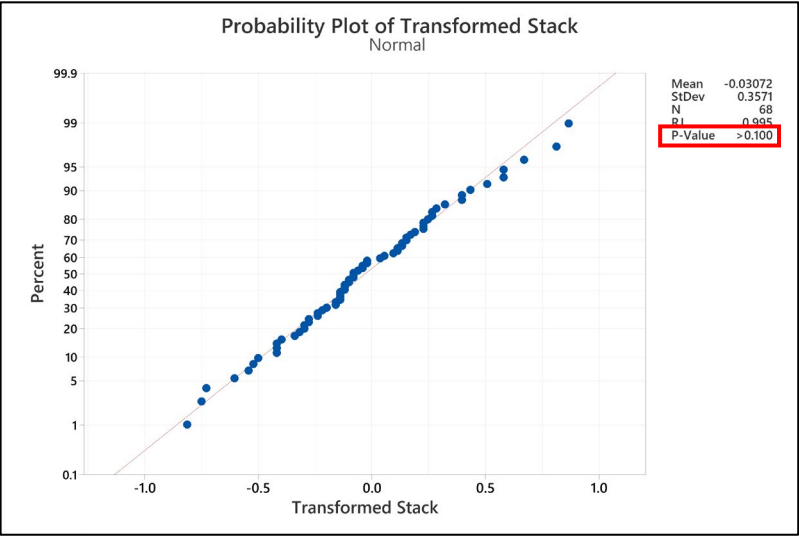
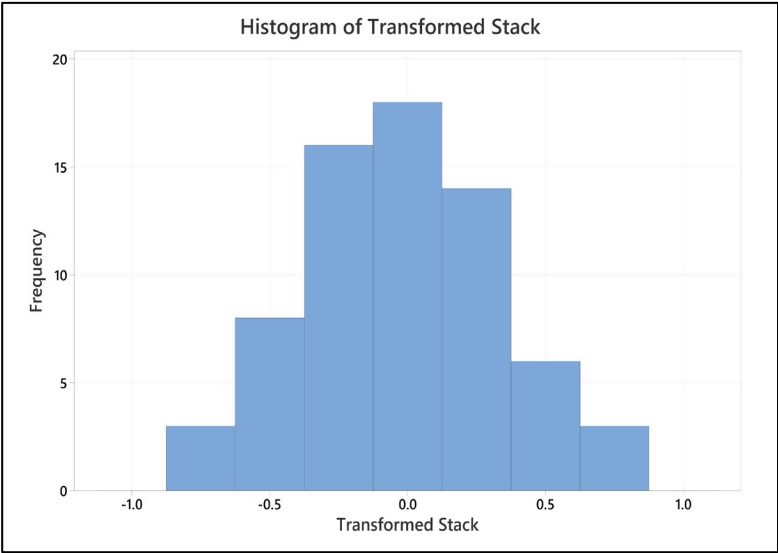
6) Let's transform the data using the Box-Cox transformation:

Stats → Control Charts → Box-Cox Transformation → All observations for a chart are in one column → Set the subgroup sizes at 4

→ Options → select “Other (enter a value between -5 and 5)” and enter 0 as the lambda value. Store the transformed stack data in a new column.



7) Check normality on the transformed data:

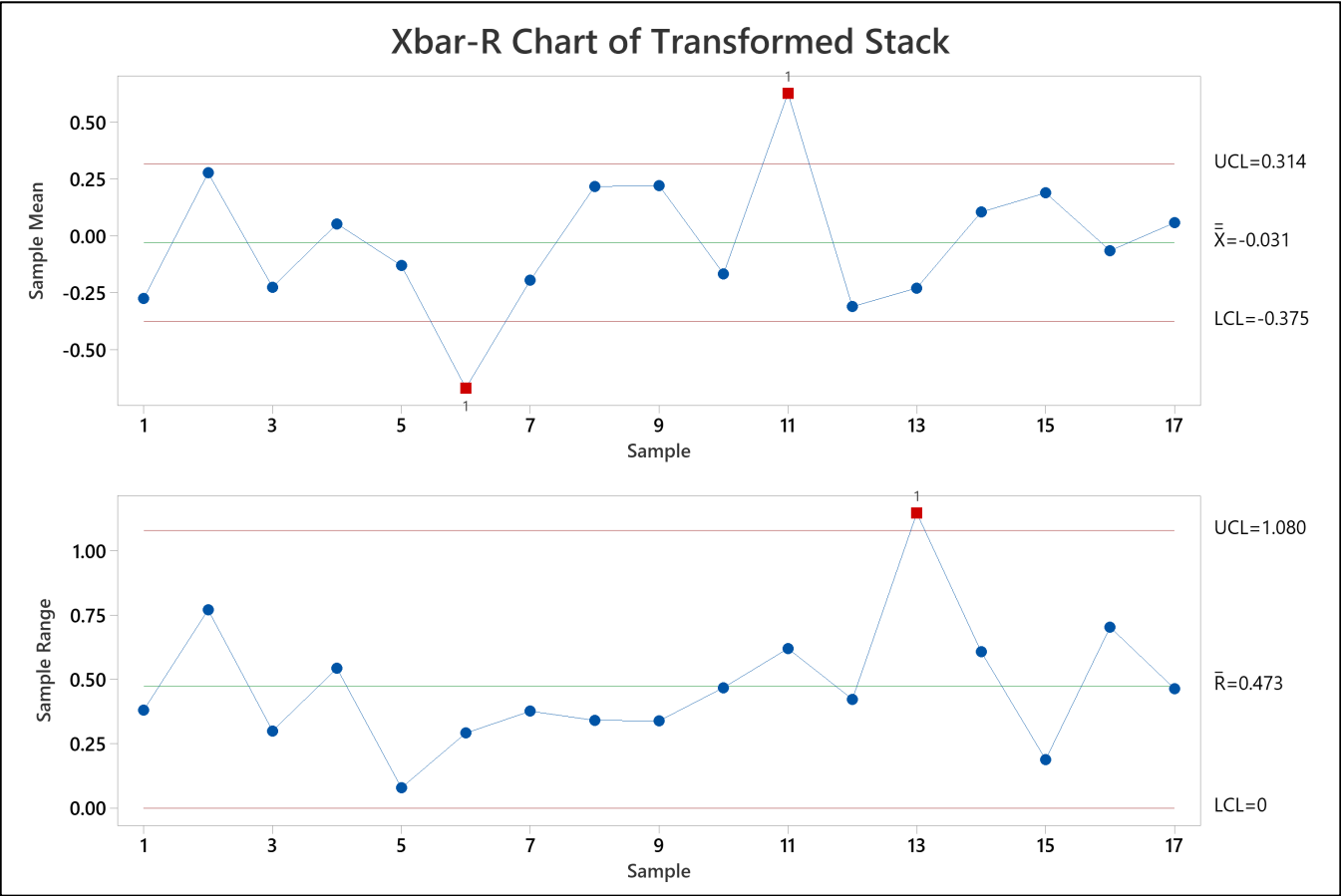


Normality is verified. We can now use the X-bar and R chart on the transformed data.

Exercise 1.1 Design a control chart for the mean.

8) Xbar-R chart on the transformed data:

*Stats → Control Charts → Variable Charts for Subgroups → Xbar-R → All observations for a chart are in one column
→ Set the Subgroup sizes to 4, or input the row subscripts column*



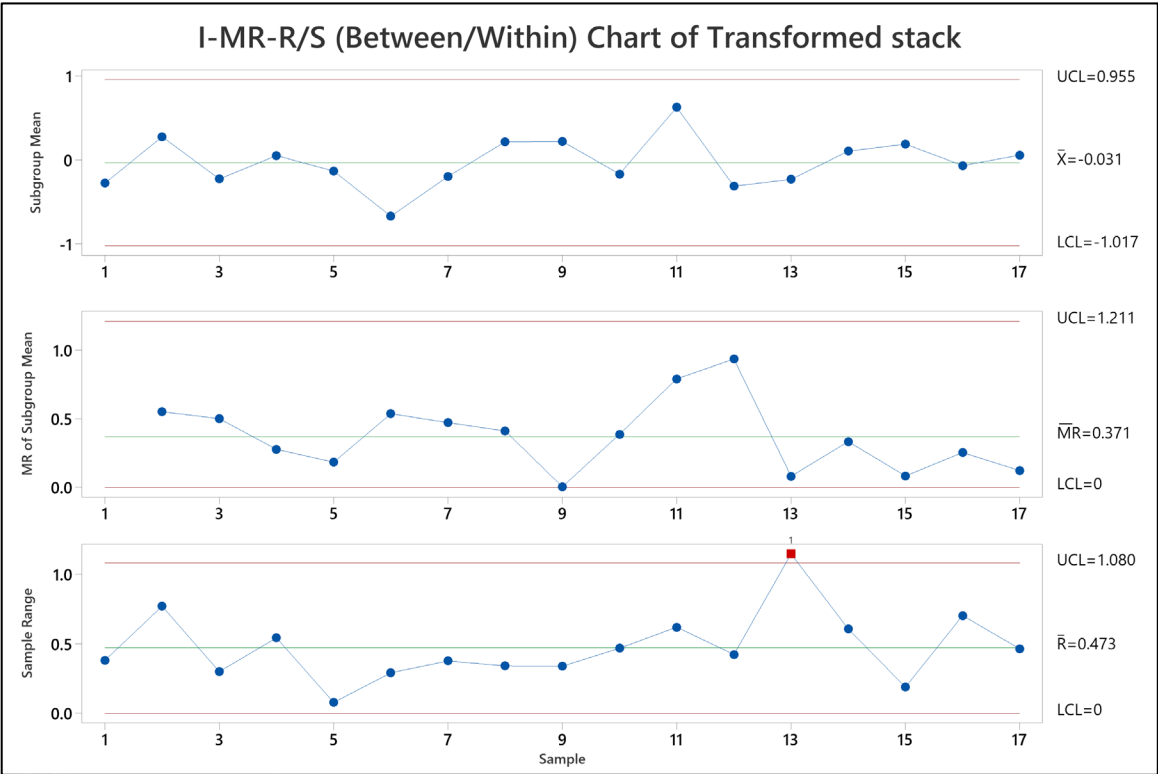
Even if the normality assumption holds, the control limits of the Xbar chart look too narrow with respect to the natural variability of the statistic. This can be caused by a violation of assumptions (independence) within the sample. Thus, the Xbar-R control chart may be not appropriate to monitor these data.

Exercise 1.2 Consider the samples as individual measurements. Design a control chart for the mean.

9) Let's design a "Between groups" control chart:

The “Between groups” control chart assumes all the samples to be individual measurements. With the I chart we can get rid of the violation of the independence assumption within the sample. The MR chart allows monitoring the between sample variability, while the R chart designed before can still be used to monitor the within sample variability. A control charting scheme quite effective in this case is the so-called I-MR-R control chart (or I-MR-S, if the S chart is used in place of the R chart), where: 1) the I chart allows monitoring the mean of the process treating sample means as individual observations; 2) the MR chart allows monitoring the between sample variability; 3) the R chart allows monitoring the within sample variability.

Stats → Control Charts → Variable Charts for Subgroups → I-MR-R/S (Between/Within) → All observations for a chart are in one column → Set the Subgroup sizes to 4, or input the row subscripts column



There is an alarm in the 13th sample. A search for assignable causes shall be performed. In the absence of information about possible assignable causes, the alarm can be labelled as a false alarm, and the design phase of the control chart is over.

Exercise 2

The data in the `LPBF.csv` file represents the deviation from the nominal shape (in mm²) of the area of 300 consecutive layers printed using laser powder bed fusion (LPBF) technology.

1. Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.
2. Design an I-MR control chart. Assume an assignable cause is found for any out-of-control points.
3. Re-design the I-MR control chart with probability limits (i.e., use the true distribution of both statistics) with $\alpha=0.012$. Regarding the MR chart, use the Box-Cox data transformation. Assume an assignable cause is found for any out-of-control points.

Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

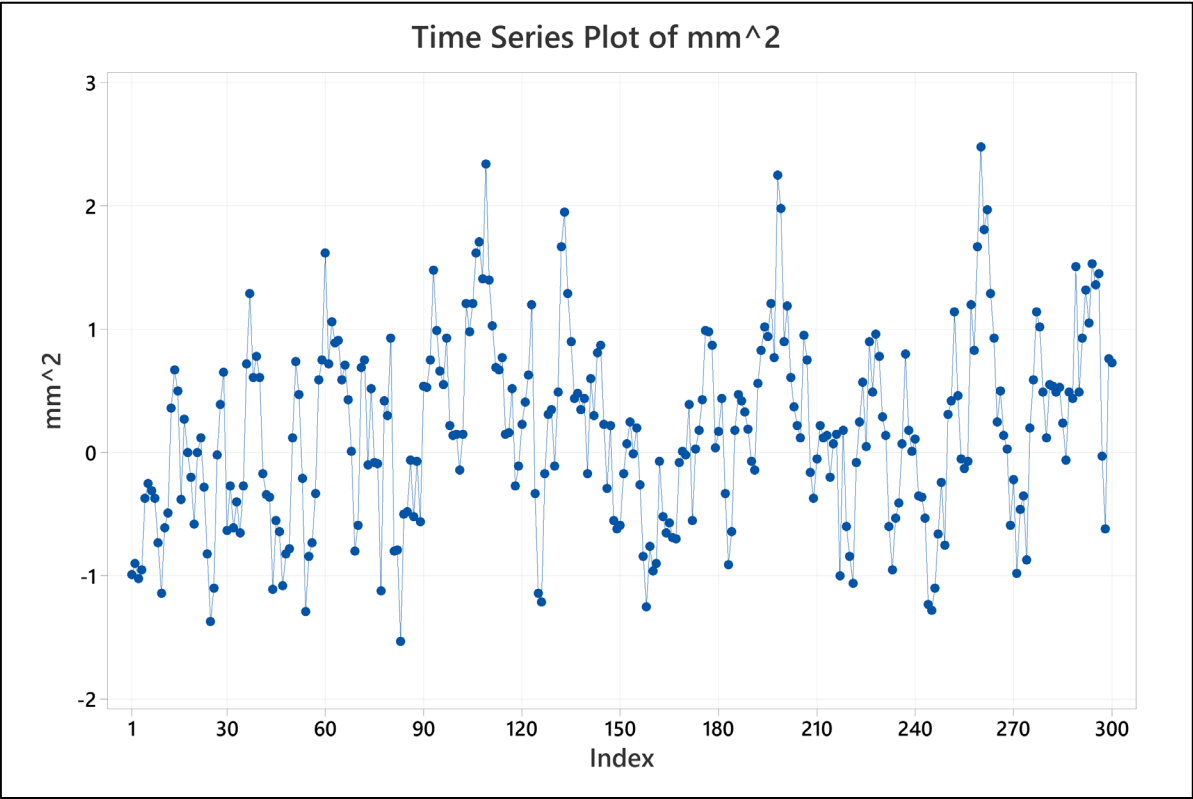
1) Import the data in Minitab:

File → Open → Locate “LPBF.csv” and Open

2) Inspect the data:

First, let’s create a time series plot:

Graph → Time Series Plot → Simple



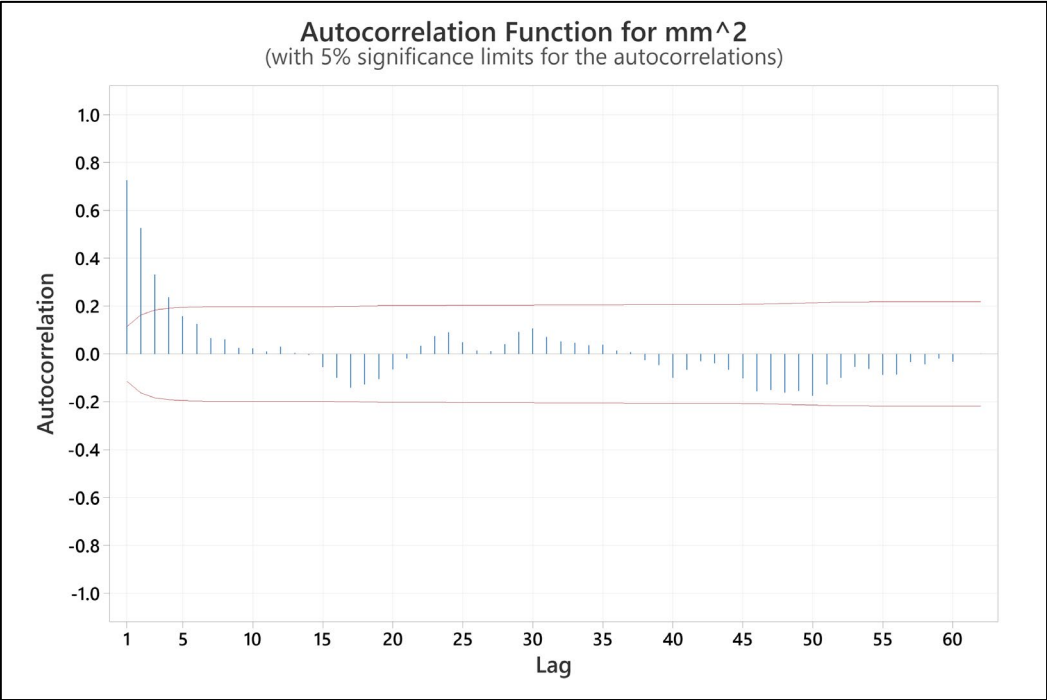
Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

1) Import the data in Minitab:

File → Open → Locate “LPBF.csv” and Open

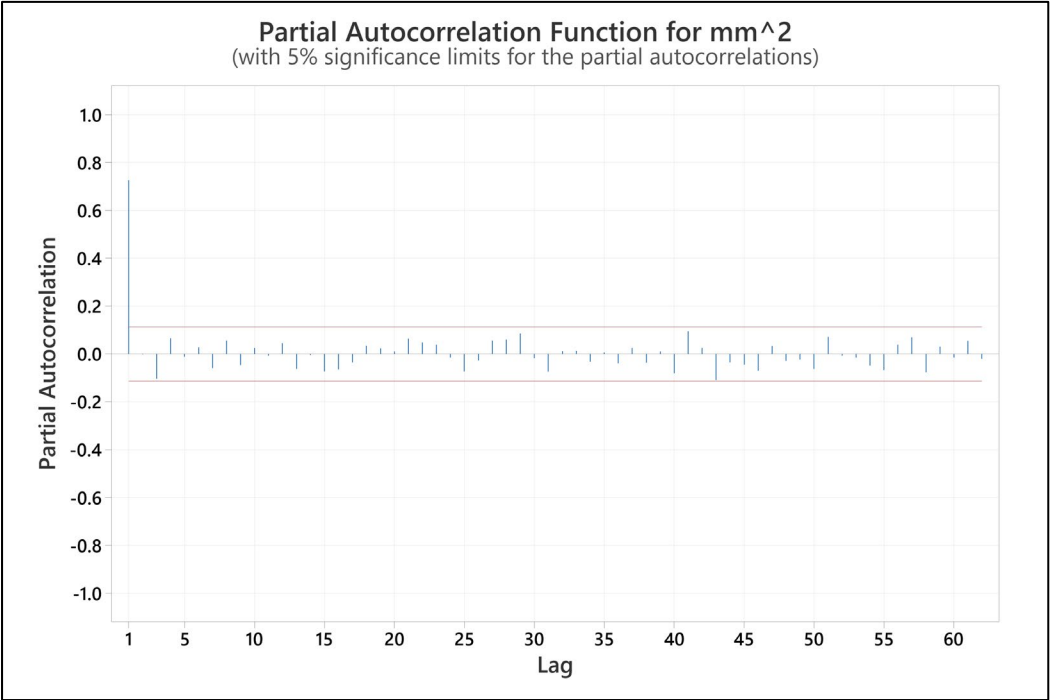
2) Inspect the data:

Let’s also plot the Autocorrelation Function (ACF):
Stat → Time Series → Autocorrelation...



Autocorrelations			
Lag	ACF	T	LBQ
1	0.725938	12.57	159.68
2	0.526376	6.36	243.92
3	0.332380	3.56	277.62
4	0.235519	2.43	294.60
5	0.157795	1.59	302.25
6	0.125124	1.25	307.07
7	0.067425	0.67	308.48
8	0.060917	0.61	309.63
9	0.024717	0.25	309.82
10	0.023954	0.24	310.00

And the Partial Autocorrelation Function (PACF):
Stat → Time Series → Partial Autocorrelation...



Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

3) Hypothesis testing (significance level 5%):

Runs test:

Stat → Nonparametrics → Runs Test...

Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
72	150.99	0.000

We reject the null hypothesis. There is statistically significant evidence to suggest that data are not random.

Bartlett’s test at lag 1:

Stat → Time Series → Autocorrelation...

BARTLETT’s TEST
for a specific lag

$H_0 : \rho_k = 0$ $H_1 : \rho_k \neq 0$

Autocorrelations			
Lag	ACF	T	LBQ
1	0.725938	12.57	159.68
2	0.526376	6.36	243.92
3	0.332380	3.56	277.62
4	0.235519	2.43	294.60
5	0.157795	1.59	302.25
6	0.125124	1.25	307.07
7	0.067425	0.67	308.48
8	0.060917	0.61	309.63
9	0.024717	0.25	309.82
10	0.023954	0.24	310.00

Critical (rejection) region:

$|r_k| > \frac{Z_{\alpha/2}}{\sqrt{n}} = 0,113$
n is 300

We reject the null hypothesis. There is statistically significant evidence to suggest that autocorrelation at lag 1 is different from 0.

Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

3) Hypothesis testing (significance level 5%):

LBQ's test at lag 4:

Stat → Time Series → Autocorrelation...

Ljung-Box (LBQ) TEST
for a global test

$H_0 : \rho_k = 0, k = 1, \dots, L$ $H_1 : \exists k \in [1, L] / \rho_k \neq 0$

Autocorrelations			
Lag	ACF	T	LBQ
1	0.725938	12.57	159.68
2	0.526376	6.36	243.92
3	0.332380	3.56	277.62
4	0.235519	2.43	294.60
5	0.157795	1.59	302.25
6	0.125124	1.25	307.07
7	0.067425	0.67	308.48
8	0.060917	0.61	309.63
9	0.024717	0.25	309.82
10	0.023954	0.24	310.00

$LBQ \sim \chi^2_L \Rightarrow$ rejection region : $LBQ > \chi^2_{\alpha,L}$

Inverse Cumulative Distribution Function:

Chi-Square with 4 DF

P(X ≤ x)	x
0.95	9.48773

Calc → Probability Distributions → Chi Squared

We reject the null hypothesis. There is statistically significant evidence to suggest that autocorrelation is different from 0 across the first 4 lags.

Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

4) Get rid of the data autocorrelation – Strategy 1: Gapping

We have 300 observations. Let's test gapping by taking every 6th observation, effectively skipping 5 observations in between each selected observation.

First, create a subscripts column to identify the observations.

Calc → Make Patterned Data → Simple Set of Numbers → Set 1 as first value, 6 as last value, and repeat the sequence 50 times.

Filter the data and store it in a new worksheet:

Data → Subset Worksheet → Select Rows

C1mm^2

C2Index

How do you want to create a subset?

Use rows that match a condition

Do you want to include or exclude rows?

Include rows that match the condition

Column:

Index

Condition:

Equals set of values

Values:

(Select All)

☒1

☐2

☐3

☐4

☐5

☐6

New worksheet name:

Gapped Data

(optional)

Select

Help

OK

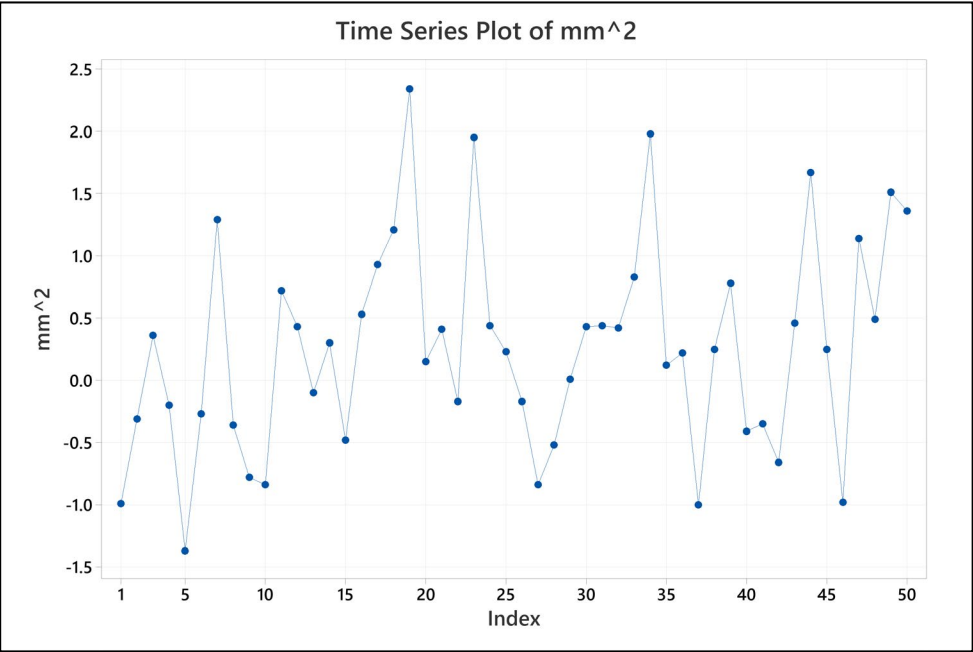
Cancel

+	C1	C2
	mm^2	Index
1	-0.99	1
2	-0.31	1
3	0.36	1
4	-0.20	1
5	-1.37	1
6	-0.27	1
7	1.29	1
8	-0.36	1
9	-0.78	1
10	-0.84	1
11	0.72	1
12	0.43	1
13	-0.10	1
14	0.30	1
15	-0.48	1
16	0.53	1
17	0.93	1
18	1.21	1
19	2.34	1
20	0.15	1
21	0.41	1
22	-0.17	1
23	1.95	1
24	0.44	1
25	0.23	1

Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

5) Get rid of the data autocorrelation – Strategy 1: Gapping

Inspect the batched data and perform hypothesis testing as before.



Test

Null hypothesis H_0 : The order of the data is random

Alternative hypothesis H_1 : The order of the data is not random

Number of Runs

Observed	Expected	P-Value
22	25.96	0.257

We reject the null hypothesis. There is statistically significant evidence to suggest that data are random.

Autocorrelations

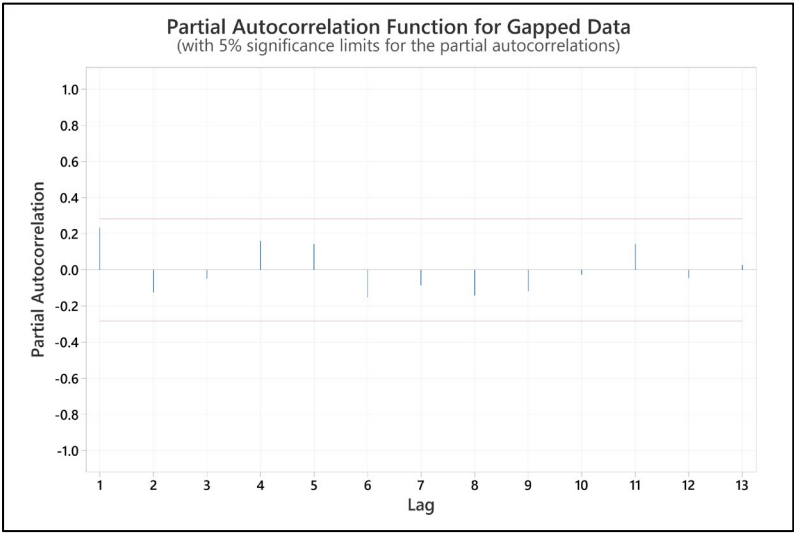
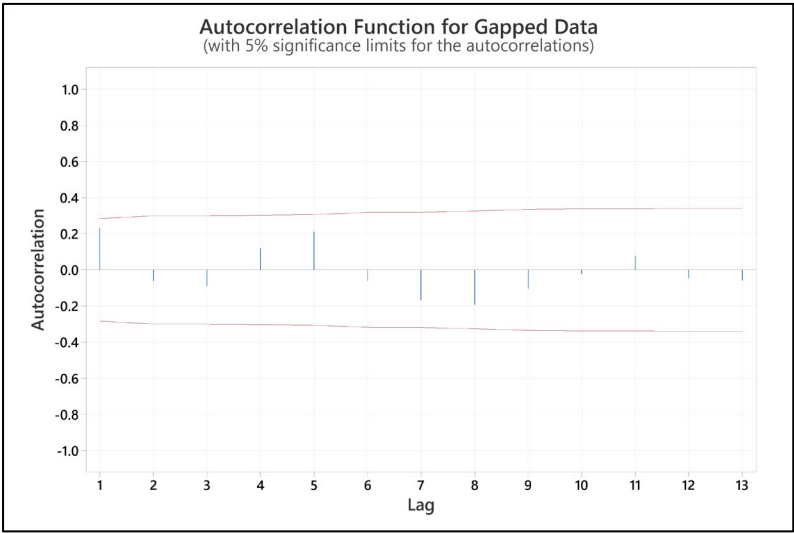
Lag	ACF	T	LBQ
1	0.233125	1.65	2.88

Bartlett’s Critical (rejection) region:

$$|r_k| > \frac{z_{\alpha/2}}{\sqrt{n}} = 0,277$$

n is now 50

Given that the lag 1 value is borderline in the ACF and PACF, we perform the Bartlett’s test. We accept the null hypothesis. There is statistically significant evidence to suggest that autocorrelation at lag 1 is 0.



Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

5) Get rid of the data autocorrelation – Strategy 2: Batching

We have 300 observations. Let’s test batching by creating 50 subgroups of 6 observations and computing their mean.

First, create a subscripts column to identify the subgroups.

Calc → Make Patterned Data → Simple Set of Numbers → Set 1 as first value, 50 as last value, and repeat each value 6 times.

Simple Set of Numbers

C1

C2

C3

C4

mm^2

Subscripts

ByVar1

Batched

Store patterned data in: C2

From first value: 1

To last value: 50

In steps of: 1

Number of times to list each value: 6

Number of times to list the sequence: 1

Select

Help

OK

Cancel

Compute the subgroup mean in a new column:

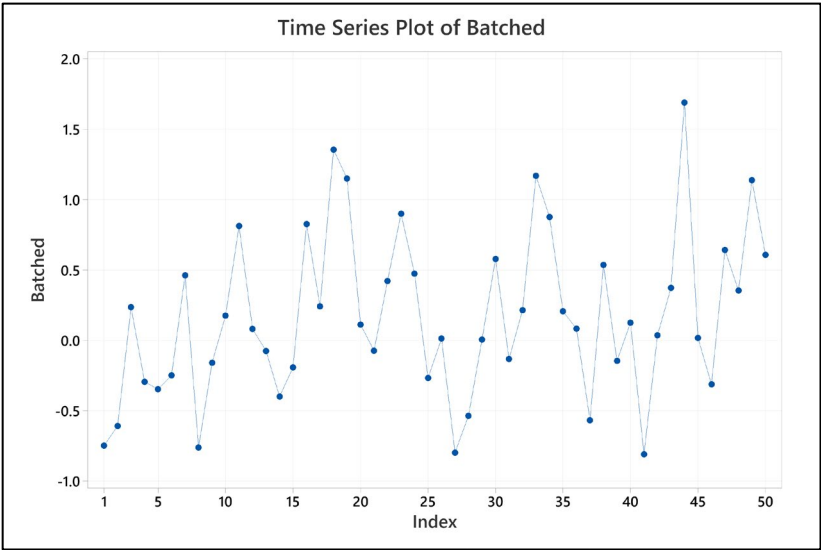
Stat → Basic statistics → Store Descriptive Statistics → Set ‘mm^2’ as variable, the subscripts for grouping by variable, and select the mean in ‘Statistics’ menu

C1	C2	C3	C4
mm^2	Subscripts	ByVar1	Batched
-0.99	1	1	-0.74667
-0.90	1	2	-0.60833
-1.02	1	3	0.23667
-0.95	1	4	-0.29333
-0.37	1	5	-0.34667
-0.25	1	6	-0.24667
-0.31	2	7	0.46333
-0.37	2	8	-0.76000
-0.73	2	9	-0.15833
-1.14	2	10	0.17667
-0.61	2	11	0.81333
-0.49	2	12	0.08167
0.36	3	13	-0.07500
0.67	3	14	-0.39833
0.50	3	15	-0.19167
-0.38	3	16	0.82667
0.27	3	17	0.24167
0.00	3	18	1.35667
-0.20	4	19	1.15000
-0.58	4	20	0.11333
0.00	4	21	-0.07333
0.12	4	22	0.42333
-0.28	4	23	0.90167

Exercise 2.1 Check if data are autocorrelated. If they are, propose a strategy to get rid of the data auto-correlation.

5) Get rid of the data autocorrelation – Strategy 2: Batching

Inspect the batched data and perform hypothesis testing as before.



Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
20	25.84	0.093

We reject the null hypothesis. There is statistically significant evidence to suggest that data are random.

Autocorrelations

Lag	ACF	T	LBQ
1	0.271556	1.92	3.91
2	-0.006990	-0.05	3.92
3	-0.115306	-0.76	4.65
4	0.070245	0.46	4.93

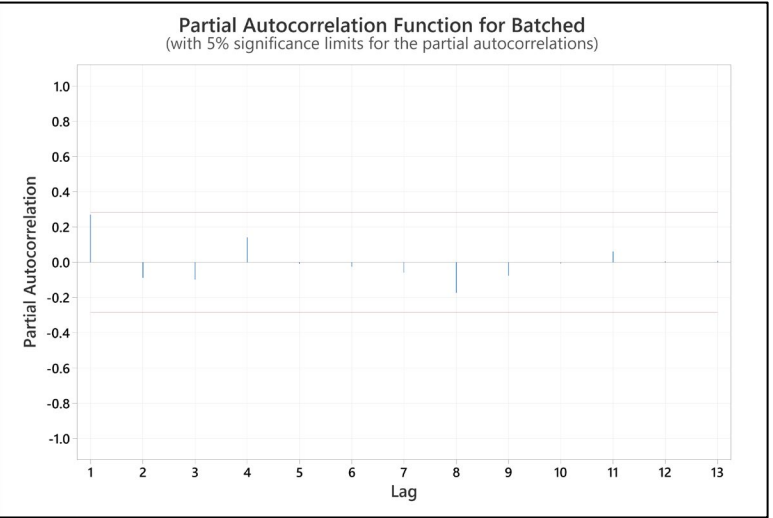
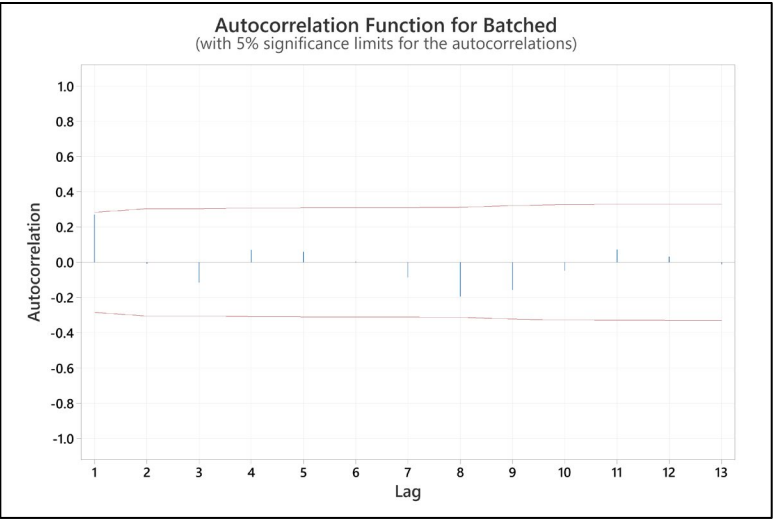
Bartlett’s Critical (rejection) region:

$$|r_k| > \frac{z_{\alpha/2}}{\sqrt{n}} = 0,277$$

n is now 50

Given that the lag 1 value is borderline in the ACF and PACF, we perform the Bartlett’s test. We accept the null hypothesis. There is statistically significant evidence to suggest that autocorrelation at lag 1 is 0.

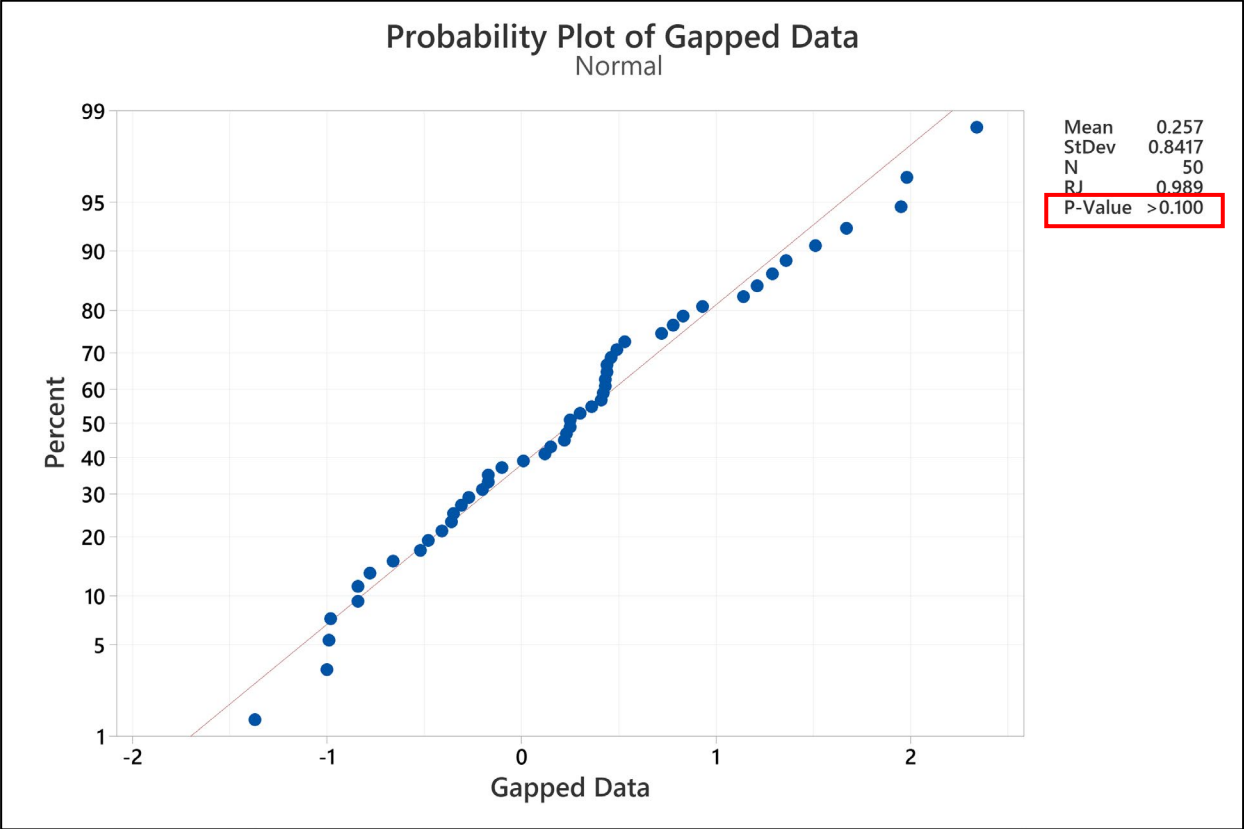
Both gapping and batching can eliminate data autocorrelation. However, based on the results of the independence tests, we have greater confidence in the independence of the data when gapping is applied. Therefore, we will proceed with the gapped data.



Exercise 2.2 Design an I-MR control chart. Assume an assignable cause is found for any out-of-control points.

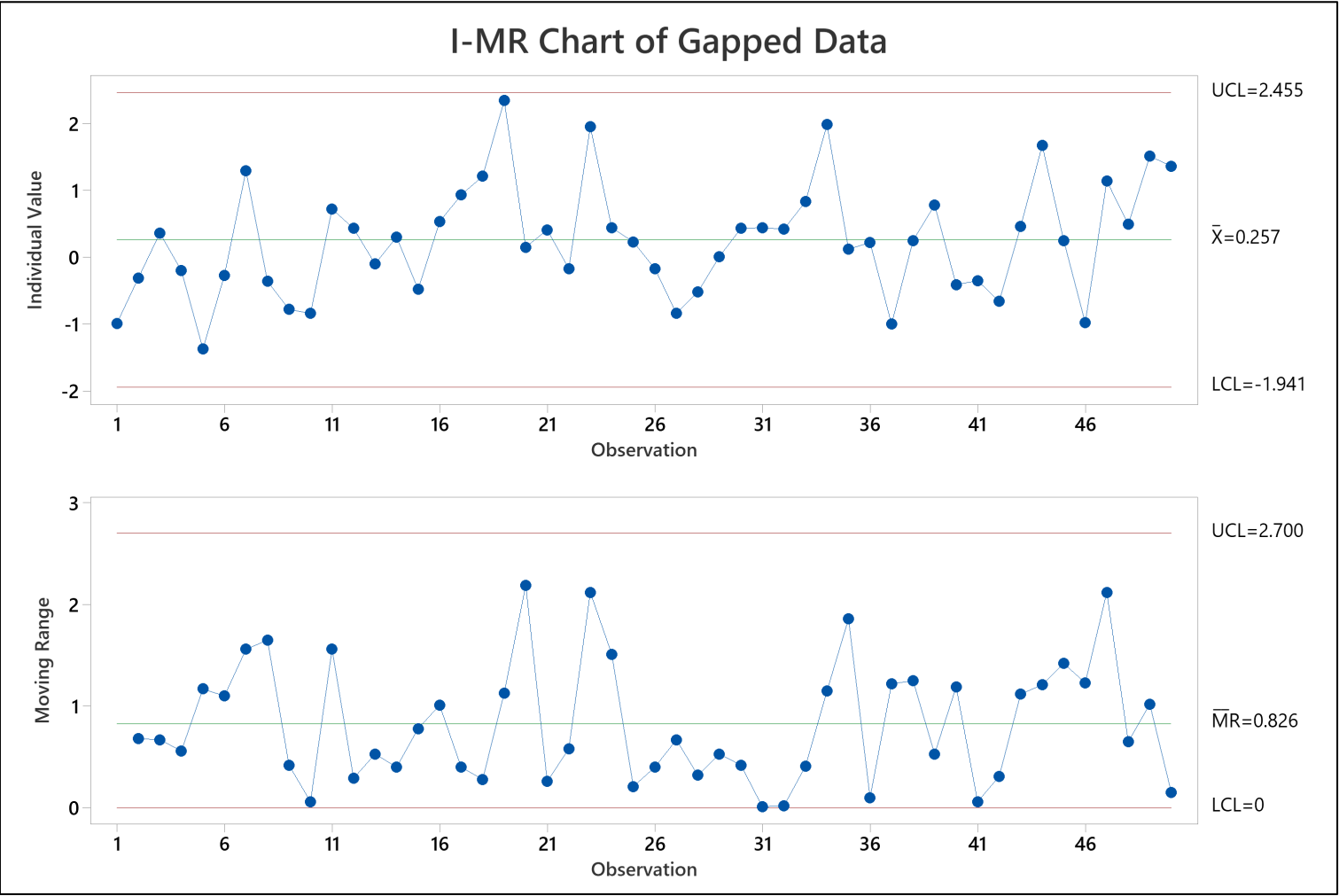
1) Check assumptions.

We have already checked independence of the gapped data.
Let's check normality.



Exercise 2.2 Design an I-MR control chart. Assume an assignable cause is found for any out-of-control points.

2) Design the I-MR chart for the gapped data.



The process is in-control.

Exercise 2.3 Re-design the I-MR control chart with probability limits (i.e., use the true distribution of both statistics) with $\alpha=0.012$. Regarding the MR chart, use the Box-Cox data transformation. Assume an assignable cause is found for any out-of-control points.

1) Compute the Type I error (α):

$$K = |z_{\alpha/2}|$$

To compute K from the Inverse cumulative normal:

*Calc → Probability Distributions
→ Normal*

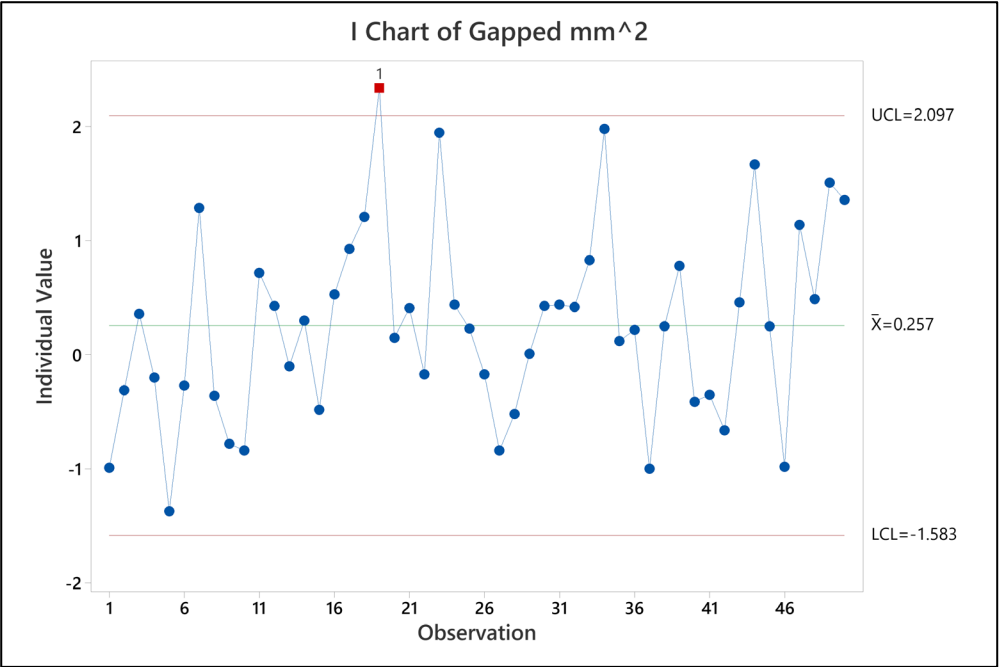
Normal with mean = 0 and standard deviation = 1

$P(X \leq x)$	x
0.006	-2.51214

2) Compute the new I chart:

*Stats → Control Charts → Variable Charts for
Individuals → Individuals*

→ I-Chart Options → Tests → $K = 2.512$



The I chart signals one out-of-control observation. Before recomputing control limits, let's check the MR chart.

Exercise 2.3 Re-design the I-MR control chart with probability limits (i.e., use the true distribution of both statistics) with $\alpha=0.012$. Regarding the MR chart, use the Box-Cox data transformation. Assume an assignable cause is found for any out-of-control points.

3) Let's design the new MR chart.

First, we must compute the MR statistic.

Calc → Calculator

C1Gapped Data

C2Index

C3MR

Store result in variable: 'MR'

Expression:
ABS(C1-LAG(C1,1))

789+ = ≠

456- < >

123* ≤ ≥

0.[] / And

^ Or

() Not

Functions:
All functions
Absolute value
Antilog
Any
Arccosh
Arccosine
Arcsine
Arcsinh

Select

☐ Assign as a formula

Help

OK

Cancel

3) Let's design the new MR chart.

We can apply a known transformation (lambda = 0.4) to the MR values and then compute probabilistic control limits.

Stats → Control Charts → Box-Cox Transformation

C1Gapped Data

C2Index

C3MR

All observations for a chart are in one column:

MR

Subgroup sizes: 1 (enter a number or ID column)

Select

Options...

Help

OK

Cancel

C1Gapped Data

C2Index

C3MR

☐ Optimal or rounded λ

☒ Other (enter a value between -5 and 5):
0.4

Store transformed data in:
C4

Select

Help

OK

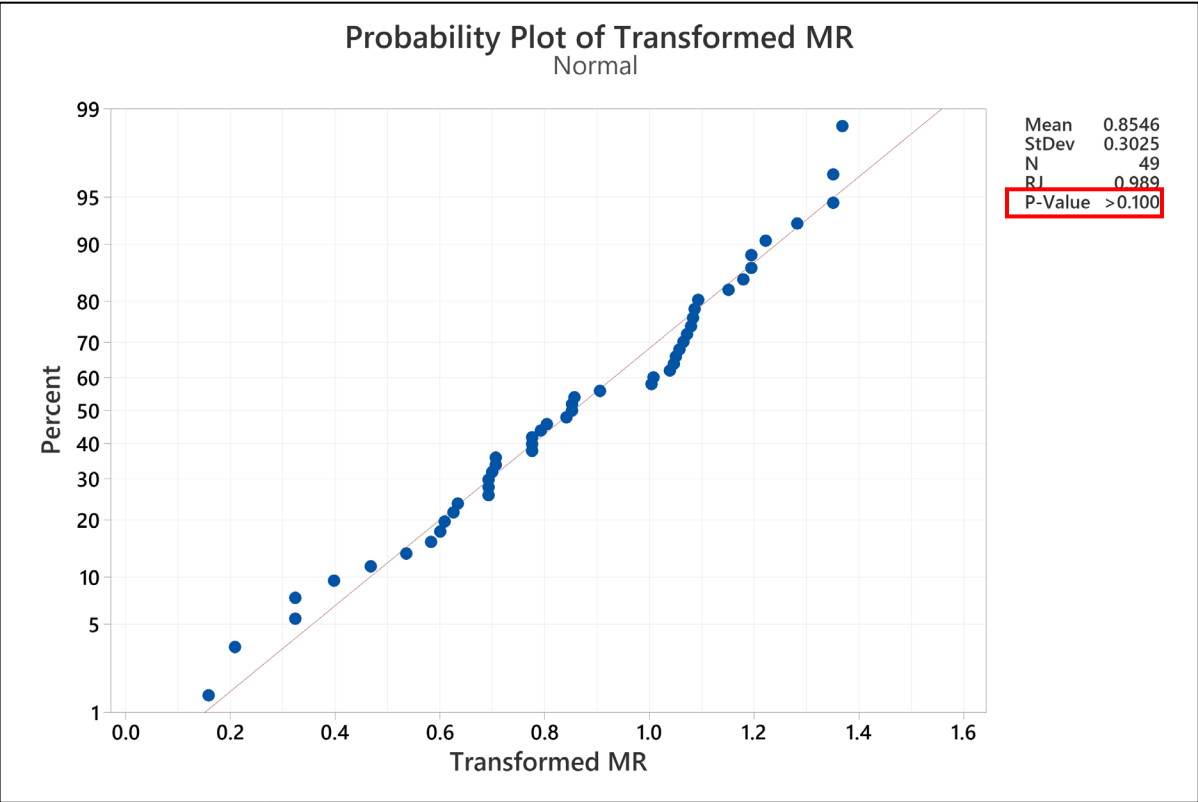
Cancel

Exercise 2.3 Re-design the I-MR control chart with probability limits (i.e., use the true distribution of both statistics) with $\alpha=0.012$. Regarding the MR chart, use the Box-Cox data transformation. Assume an assignable cause is found for any out-of-control points.

3) Let's design the new MR chart.

Let's check if the transformed MR statistic is now normal.

Stat → Basic Statistics → Normality Test → Shapiro-Wilk test

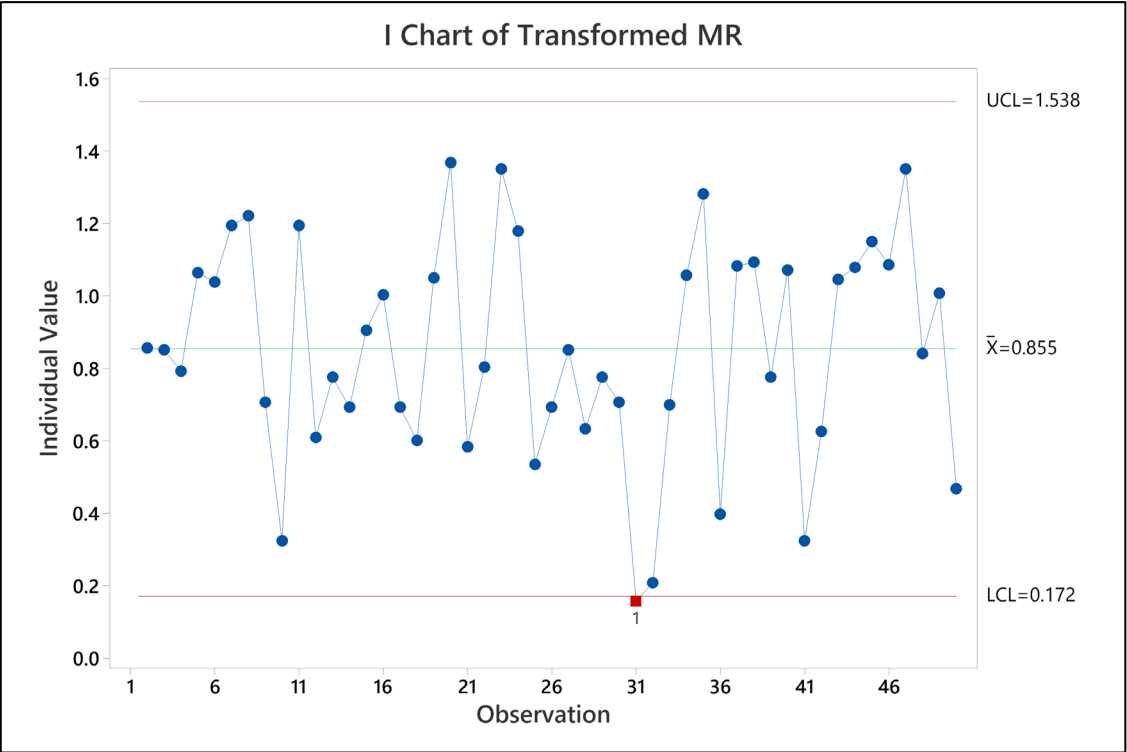


3) Let's design the new MR chart.

Let's compute the I chart with probabilistic control limits on the transformed MR data.

Stats → Control Charts → Variable Charts for Individuals → Individuals

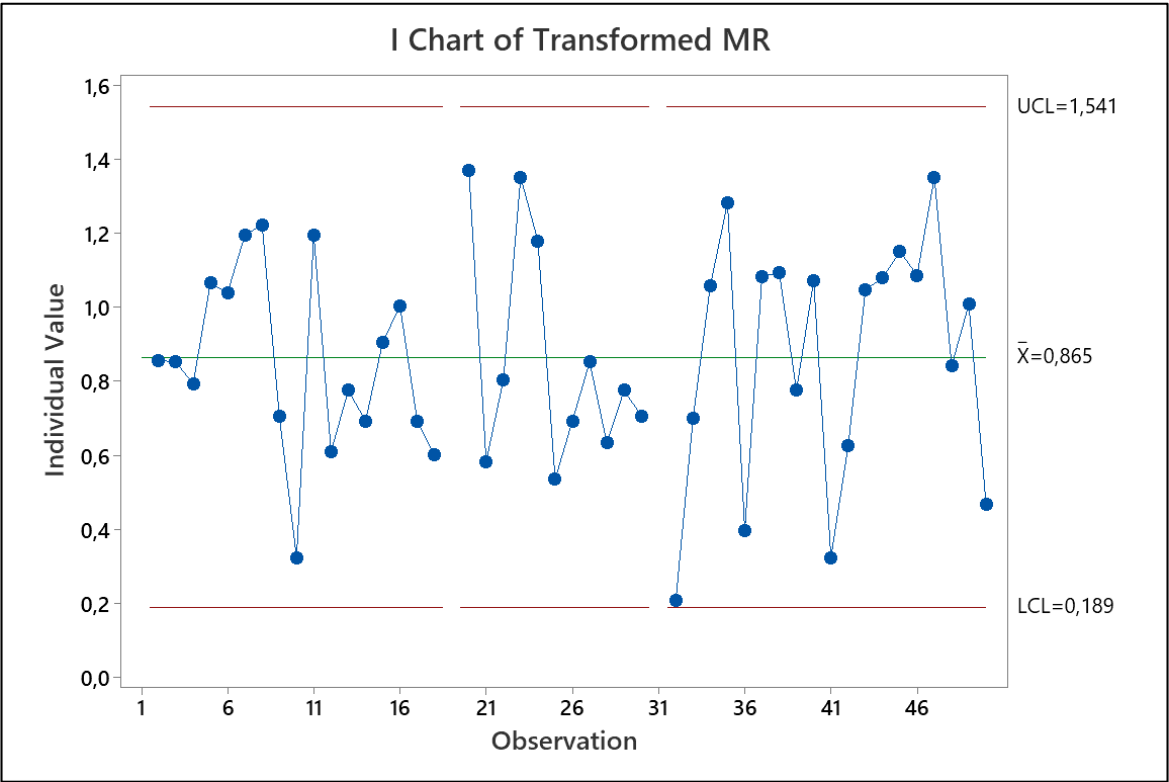
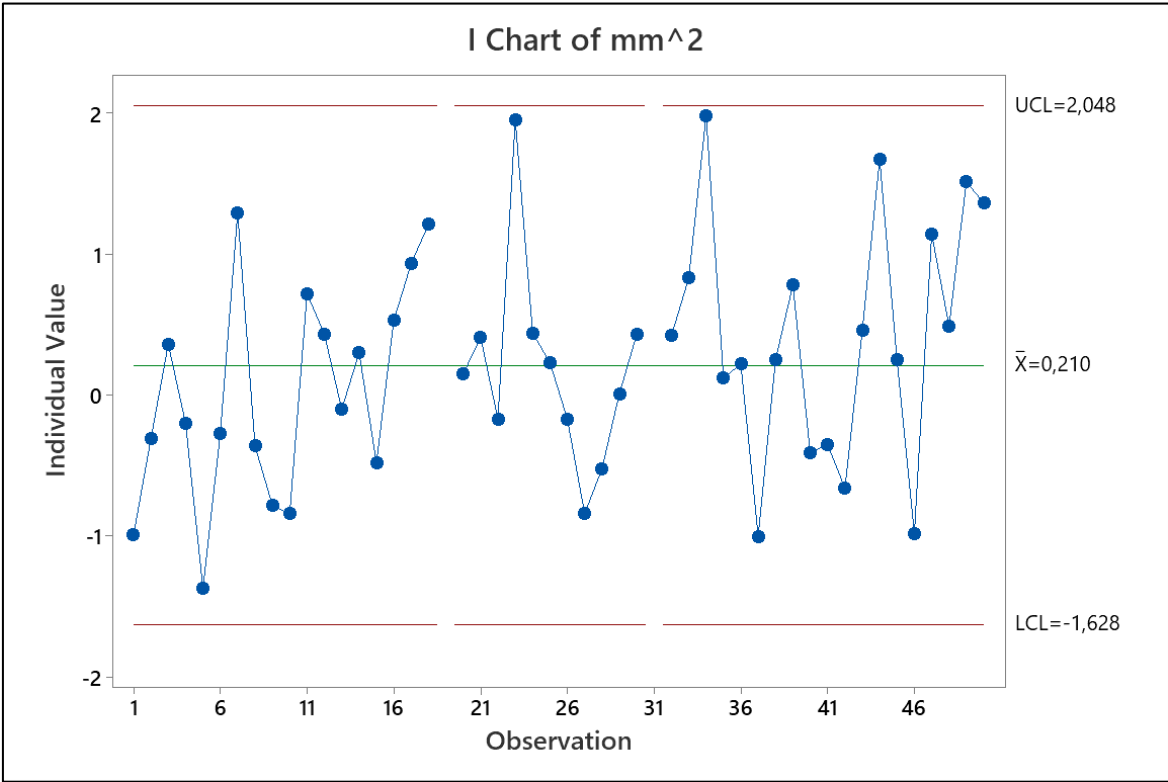
→ I-Chart Options → Tests → K = 2.512



Exercise 2.3 Re-design the I-MR control chart with probability limits (i.e., use the true distribution of both statistics) with $\alpha=0.012$. Regarding the MR chart, use the Box-Cox data transformation. Assume an assignable cause is found for any out-of-control points.

4) Recompute control limits for the I and MR charts.

We must remove the out-of-control observations and recompute control limits. We must remove observations 19 and 31 from gapped data.



The process is now in-control.